

to maximize a government's revenues—and we'll see how to use an algorithm to find the right solution.

Setting Tax Rates

Imagine that you're elected prime minister of a small country. You have ambitious goals, but you don't feel like you have the budget to achieve them. So your first order of business after taking office is to maximize the tax revenues your government brings in.

It's not obvious what taxation rate you should choose to maximize revenues. If your tax rate is 0 percent, you will get zero revenue. At 100 percent, it seems likely that taxpayers would avoid productive activity and assiduously seek tax shelters to the point that revenue would be quite close to zero. Optimizing your revenue will require finding the right balance between rates that are so high that they discourage productive activity and rates that are so low that they undercollect. To achieve that balance is, you'll need to know more about the way tax rates relate to revenue.

Steps in the Right Direction

Suppose that you discuss this with your team of economists. They see your point and retire to their research office, where they consult the apparatuses used by top-level research economists everywhere—mostly test tubes, hamsters running on wheels, astrolabes, and dowsing rods—to determine the precise relationship between tax rates and revenues.

After some time thus sequestered, the team tells you that they've determined a function that relates the taxation rate to the revenue collected, and they've been kind enough to write it in Python for you. Maybe the function looks like the following:

```
import math
def revenue(tax):
    return(100 * (math.log(tax+1) - (tax - 0.2)**2 + 0.04))
```

This is a Python function that takes `tax` as its argument and returns a numeric output. The function itself is stored in a variable called `revenue`. You